HPE ARUBA OS-CX CLI CHEAT SHEET

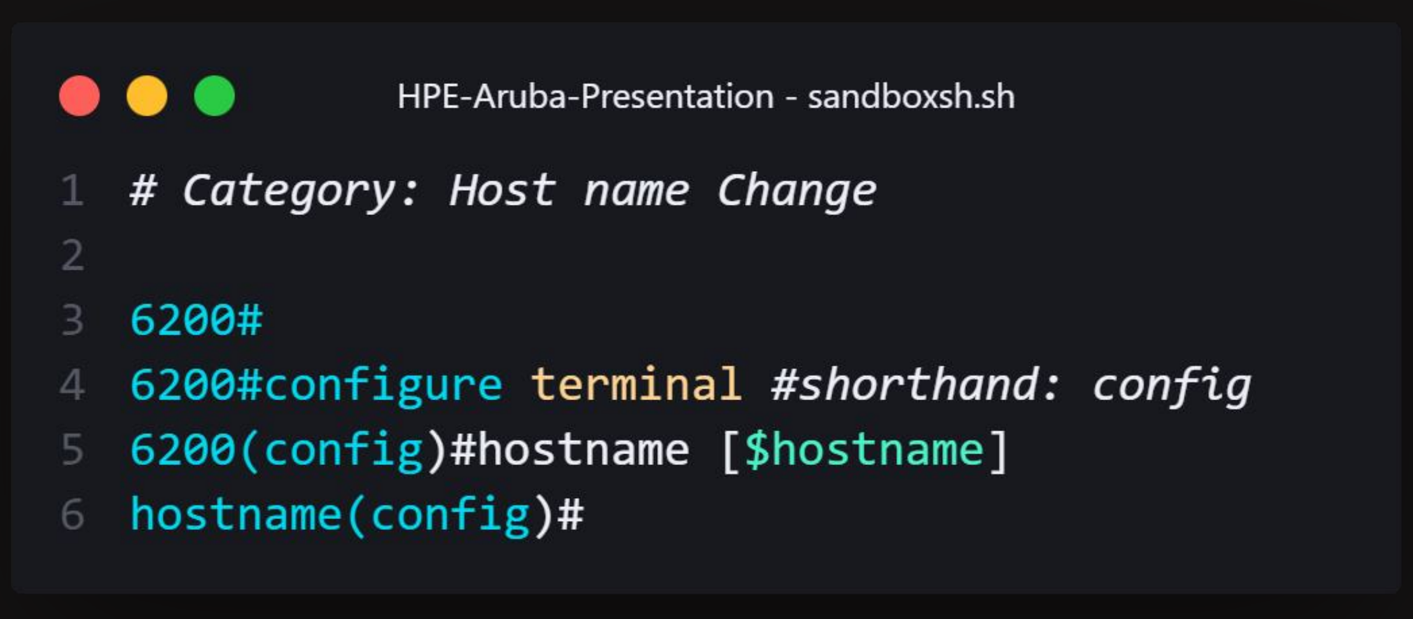
Essential Commands for Basic Switch Setup and Operations

An essential CLI guide to set up HPE Aruba switches to an operating state, from basic security to VLAN and routing configuration.

Step 1: Hostname Change

What is it for?

Changing a hostname replaces a network device's generic factory name with a unique, descriptive identifier, which is essential for distinguishing individual switches during remote management, ensuring accurate system logging and monitoring, and preventing administrators from accidentally making configuration changes on the wrong device.



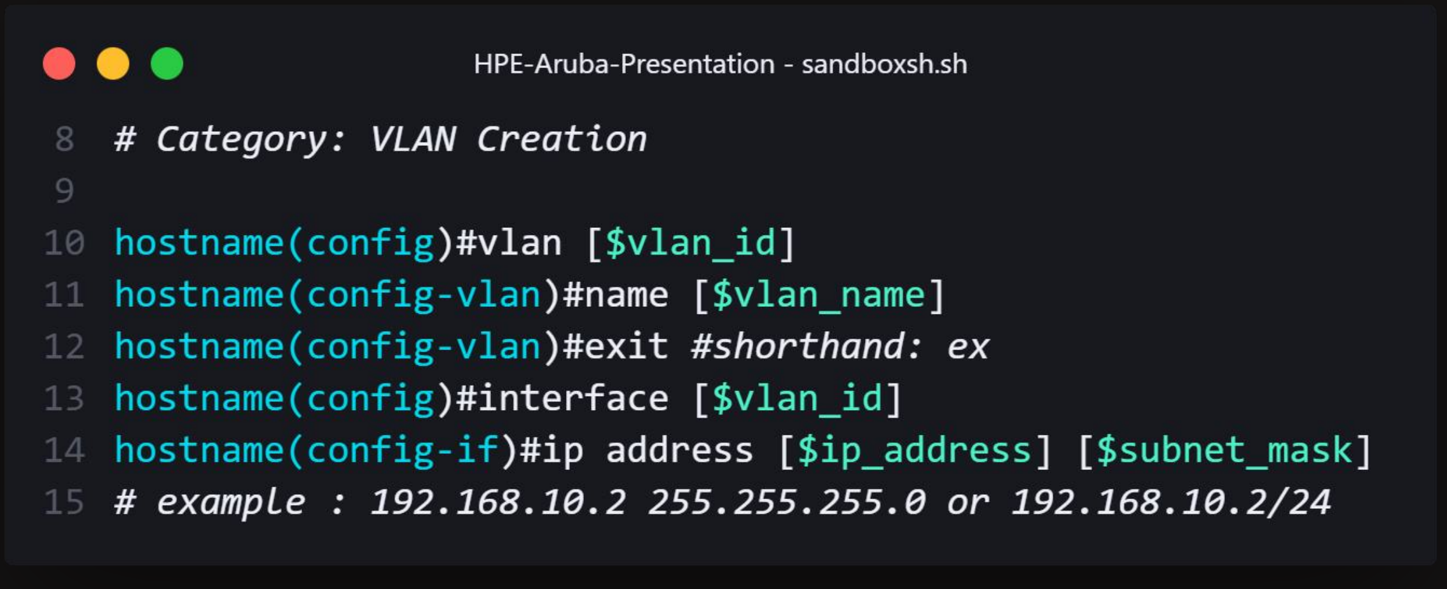
```
# Category: Host name Change

6200#
6200#configure terminal #shorthand: config
6200(config)#hostname [$hostname]
hostname(config)#
```

Step 2: VLAN Creation & SVI (Interface) Configuration

What is it for?

Traffic Isolation & Performance: VLANs break large enterprise networks into smaller, isolated broadcast domains (e.g., separating Voice, Guest, and Corporate traffic), reducing network congestion caused by broadcast storms.



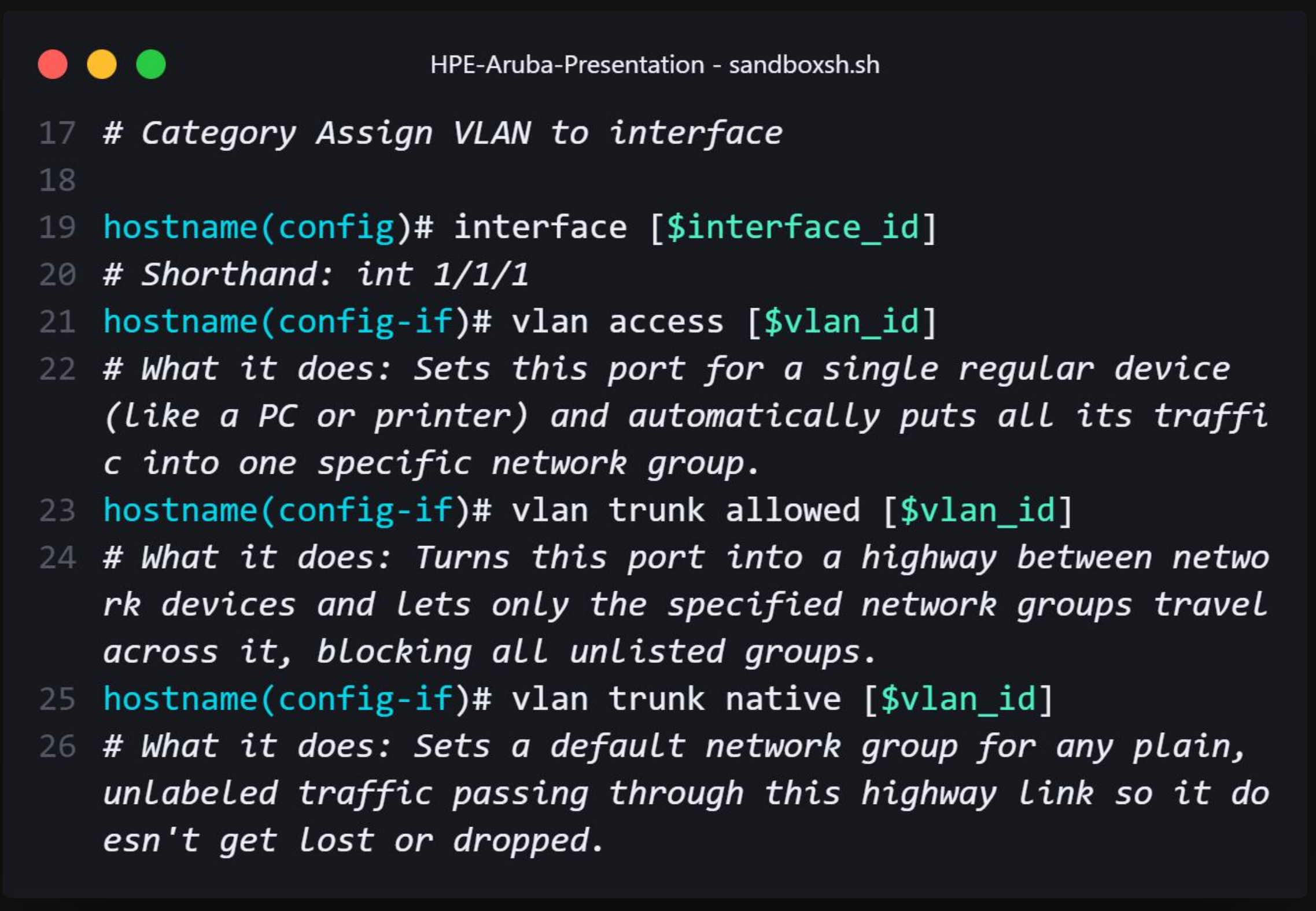
```
# Category: VLAN Creation

hostname(config)#vlan [$vlan_id]
hostname(config-vlan)#name [$vlan_name]
hostname(config-vlan)#exit #shorthand: ex
hostname(config)#interface [$vlan_id]
hostname(config-if)#ip address [$ip_address] [$subnet_mask]
# example : 192.168.10.2 255.255.255.0 or 192.168.10.2/24
```

Step 3: Assigning VLANs to Switch Interfaces

What is it for?

VLAN assignment configures physical switch ports to control how network traffic is handled at Layer 2: an access port links to an individual endpoint (like a PC) and assigns all incoming untagged traffic to a single specific VLAN; a trunk allowed configuration permits traffic from multiple tagged VLANs to pass through a single switch-to-switch link; and a trunk native setting identifies the designated single VLAN that receives any untagged traffic passing across that trunk link without dropping it



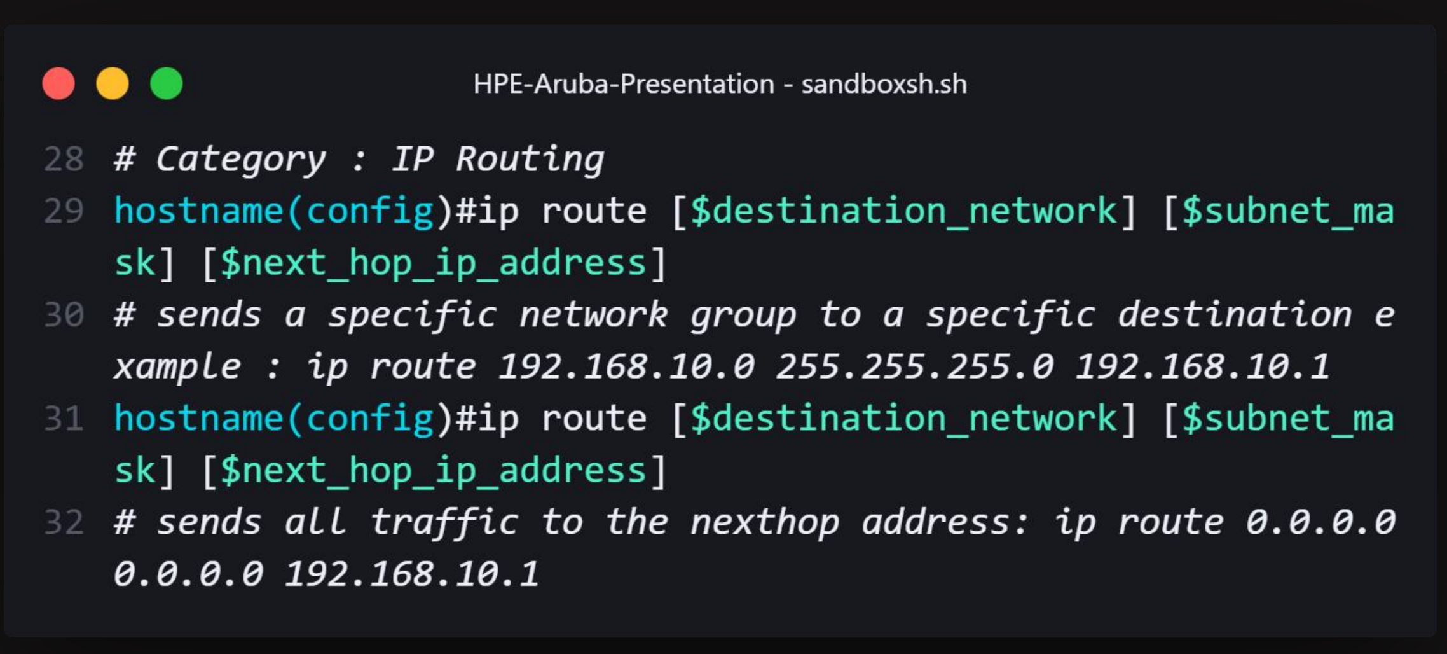
```
# Category Assign VLAN to interface

hostname(config)# interface [$interface_id]
# Shorthand: int 1/1/1
hostname(config-if)# vlan access [$vlan_id]
# What it does: Sets this port for a single regular device (like a PC or printer) and automatically puts all its traffic into one specific network group.
hostname(config-if)# vlan trunk allowed [$vlan_id]
# What it does: Turns this port into a highway between network devices and lets only the specified network groups travel across it, blocking all unlisted groups.
hostname(config-if)# vlan trunk native [$vlan_id]
# What it does: Sets a default network group for any plain, unlabeled traffic passing through this highway link so it doesn't get lost or dropped.
```

Step 4: Assigning VLANs to Switch Interfaces

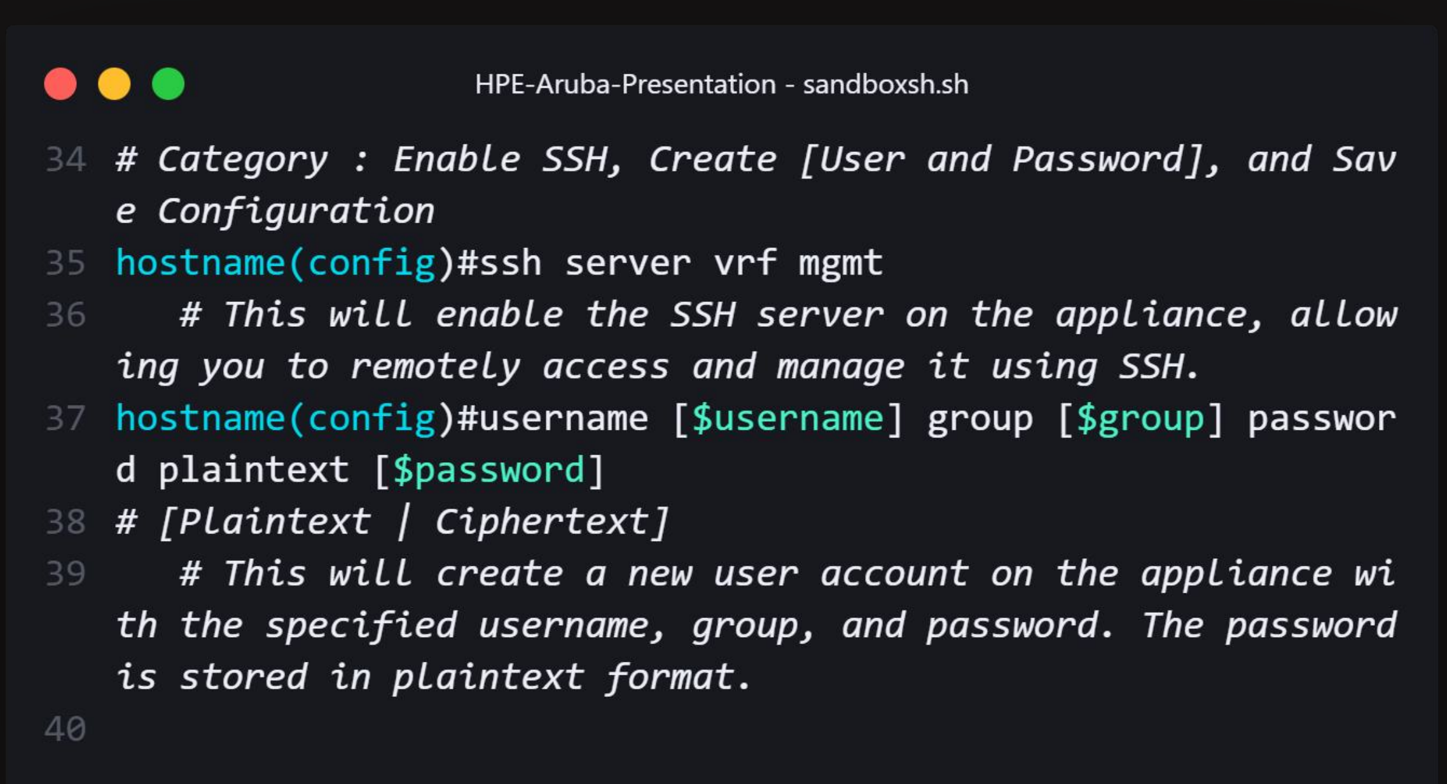
What is it for?

IP routing is the process in networking that enables data packets to travel between different subnets and distinct broadcast domains by examining each packet's destination IP address and matching it against a routing table to select the optimal next-hop path toward its final destination or default gateway.



```
# Category : IP Routing
hostname(config)#ip route [$destination_network] [$subnet_mask] [$next_hop_ip_address]
# sends a specific network group to a specific destination example : ip route 192.168.10.0 255.255.255.0 192.168.10.1
hostname(config)#ip route [$destination_network] [$subnet_mask] [$next_hop_ip_address]
# sends all traffic to the nexthop address: ip route 0.0.0.0 0.0.0.0 192.168.10.1
```

Step 5: Basic Security & User Access Setup



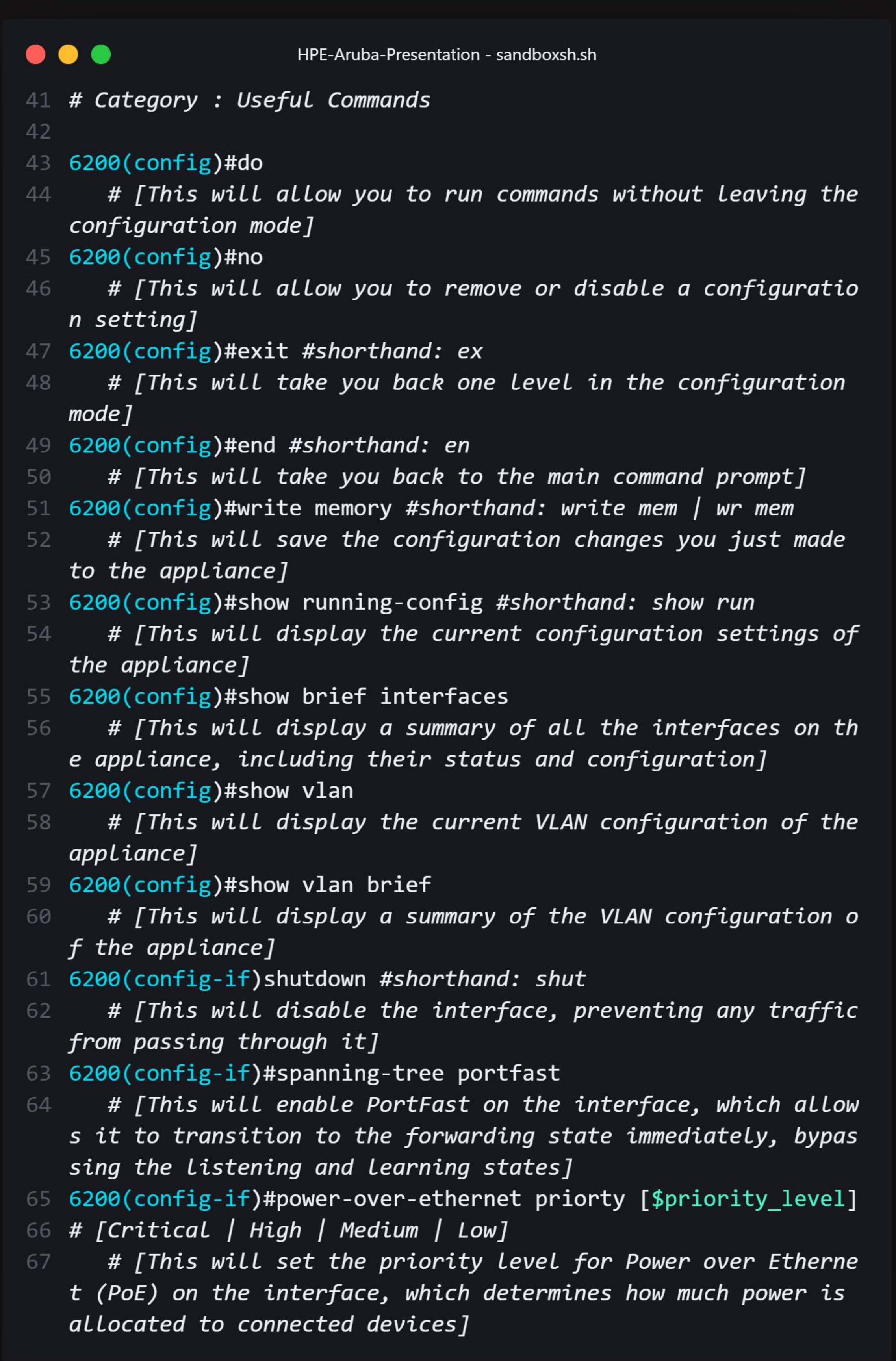
```
# Category : Enable SSH, Create [User and Password], and Save Configuration
hostname(config)#ssh server vrf mgmt
   # This will enable the SSH server on the appliance, allowing you to remotely access and manage it using SSH.
hostname(config)#username [$username] group [$group] password plaintext [$password]
# [Plaintext | Ciphertext]
   # This will create a new user account on the appliance with the specified username, group, and password. The password is stored in plaintext format.

```

What is it for?

Setting up SSH access and creating a user account establishes secure, encrypted administrative access to a network switch, replacing clear-text management protocols with protected remote management while enforcing role-based access control so that only authenticated administrators with designated privilege levels can access and configure the device.

Useful Operational & Interface Tuning Commands



```
# Category : Useful Commands

6200(config)#do
   # [This will allow you to run commands without leaving the configuration mode]
6200(config)#no
   # [This will allow you to remove or disable a configuration setting]
6200(config)#exit #shorthand: ex
   # [This will take you back one level in the configuration mode]
6200(config)#end #shorthand: en
   # [This will take you back to the main command prompt]
6200(config)#write memory #shorthand: write mem | wr mem
   # [This will save the configuration changes you just made to the appliance]
6200(config)#show running-config #shorthand: show run
   # [This will display the current configuration settings of the appliance]
6200(config)#show brief interfaces
   # [This will display a summary of all the interfaces on the appliance, including their status and configuration]
6200(config)#show vlan
   # [This will display the current VLAN configuration of the appliance]
6200(config)#show vlan brief
   # [This will display a summary of the VLAN configuration of the appliance]
6200(config-if)shutdown #shorthand: shut
   # [This will disable the interface, preventing any traffic from passing through it]
6200(config-if)#spanning-tree portfast
   # [This will enable PortFast on the interface, which allows it to transition to the forwarding state immediately, bypassing the listening and learning states]
6200(config-if)#power-over-ethernet priorty [$priority_level]
# [Critical | High | Medium | Low]
   # [This will set the priority level for Power over Ethernet (PoE) on the interface, which determines how much power is allocated to connected devices]
```

What is it for?

Saves running configurations to persistent memory using write memory.
Monitors interface and VLAN states using show brief interfaces and show vlan.
Controls port states via shutdown / no shutdown.
Optimizes Spanning Tree behavior using spanning-tree portfast.
Controls Power over Ethernet budget allocation using power-over-ethernet priority [Priority_level]

What is it for?

Volatile Memory Persistence: Commands executed in OS-CX reside in RAM (running-config); executing write memory commits them to flash storage so configurations survive power outages or reboots.

Loops & Rapid Convergence: Enabling spanning-tree portfast on edge access ports allows end-user devices to bypass the standard Spanning Tree listening and learning states, transitioning immediately to the forwarding state to prevent DHCP timeouts upon boot-up.

Resource Prioritization (PoE): Assigning PoE priority levels ensures critical devices (like security cameras or emergency IP phones) remain powered if total switch power draw exceeds available power supply limits.

HPE Aruba Networking | For official documentation and firmware updates, visit asp.arubanetworks.com

Getting Help: Type ? at any CLI prompt to view available commands and parameter options.